Indicating Ambiguous False Positives to Improve Wide-Area SAR Vessel Detection

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***Abstract*— Automatic vessel detection based on wide-area synthetic aperture radar (SAR) imagery is achieved via a statistical hypothesis test in which a ship is declared detected when the pixel intensity exceeds a predetermined threshold. Inherent to the physical principle of SAR as a pulsed radar system, not only legitimate vessels are detected but also ambiguous reflections caused by the periodic sampling of the scene. These ambiguous returns occur in both directions, azimuth and range, and can originate at land or ships. Depending on the chosen beam mode and system parameters, many also exceed the threshold and can severely degrade the overall performance quality. In the literature, several elaborate technological and signal-processing techniques were proposed to either entirely avoid the emerging of ambiguities or at least suppress them significantly. While hardware-based solutions are not readily available on existing spacecraft and potentially cost prohibitive on future ones, most processing algorithms are computationally too laborious for near real-time applications. This article presents an alternative solution. As SAR-based ship detection is not about creating neat imagery, ambiguities may be tolerated as long as they can be reliably identified as such during a follow-on examination. This step is being proposed to be a novel additional statistical test that decides which hypothesis (vessel or ambiguity probability) was more likely to have generated the prior detected pixel cluster. The effectiveness of the method is shown theoretically and corroborated with real RADARSAT Constellation Mission (RCM) data.**

***Index Terms*— Doppler ambiguities, range ambiguities, spaceborne synthetic aperture radar (SAR), vessel detection.**

I. Introduction

WIDE-AREA vessel detection has become an operational application of several space-based synthetic aperture radar (SAR) systems such as RADARSAT-2, RCM [1], and Sentinel-1 [2]. Automatic vessel detection based on single- or multipolarized SAR images is achieved through a statistical hypothesis test in which a ship detection is declared when a pixel magnitude exceeds a predetermined threshold. The brightness (SAR image intensity) is used as discriminator between vessels and the sea background commonly called clutter. Caused by the physical principle of space-based SAR not only legitimate target vessels but also many false alarms or false positives are detected, which can be caused by natural physical phenomena such as breaking waves in higher sea states but also through ambiguous reflections caused by the periodic sampling of the scene inherent to pulsed radar systems. The detection threshold is commonly chosen to adhere to a constant false alarm rate (CFAR), which is based on a statistical model for the background sea clutter and noise. The operational CFAR requirement for RCM is 2.5e−9, meaning on average that less than 2 pixels over a 10 min imaging time are expected to be false positives [1], [3]. Over several decades, the K-distribution was used as the standard model for the sea surface [4].

Compared with older C-band SARs such as RADARSAT-1/2 or ENVISAT with coarser resolutions, modern systems such as RCM and Gaofen-3 are capable of resolving individual ocean features such as sea spikes (breaking waves) that mostly appear in near range or in higher sea states. Since such spikes behave similar to strong point-like responses and not like the rest of the sea echoes, they can be falsely declared as vessels. To avoid an exceedingly large number of false alarms, the chosen statistical model must incorporate a higher likelihood of larger sea returns, i.e., must include an enlarged tail of the probability density function (pdf). To reduce these kind of false positives on the open ocean, over the years many different distributions have been proposed [5]. One promising model called tri-modal discrete (3MD) has recently proven to perform well under highly heterogeneous sea conditions commonly encountered in higher sea states [6] and is since being applied operationally on RCM [7].

However, an additional crucial source of false detections are image artifacts called ambiguities. For vessel detection using space-based SAR imagery, image quality is not the driving force. Instead beam modes are optimized to comply with two main requirements, large coverage (swath width) and detectability of ships down to a predefined size [1]. A large swath is created by a combination of large incidence angles and a low pulse repetition frequency (PRF). Although high incidence angles are, on one hand, beneficial because they reduce the return from the sea surface, they also increase the occurrence of range ambiguities as far away scatterers become illuminated by the antenna main lobe. On the other hand, a low PRF increases the chance of emerging azimuth ambiguities. Ambiguous "ghost targets" in SAR images resemble in many cases reflections from desired legitimate vessels and may not be discernible. Although the inherent causes for azimuth and range ambiguities are different, the effect is identical, i.e., if too many ambiguous ghost targets exceed the detection threshold, they can either fool or overwhelm image analysts, and in worst cases may render the ship detection tool unserviceable.

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II. Background and Existing Methods

Over the past decades, many different hardware- and software-based techniques have been proposed in the literature to rid the SAR imagery of the unfavorable artifacts caused by ambiguities.

A. Range Ambiguities

The occurrence of range ambiguities is inherent to a pulsed radar system. They are caused by echoes from scatterers that lie outside of the desired range extent, i.e., returns from previous or later transmitted pulses superimposed onto the correct echoes. Conventionally, range ambiguities are mitigated by suitably shaping the elevation antenna pattern attempting to increase sidelobe attenuation and thereby filtering out the undesired range contributions. However, to achieve this, a larger SAR antenna in elevation is required, which in most cases is cost prohibitive. If the suppression by the antenna sidelobes is not adequate, additional mitigation procedures are needed. A common practice for some systems, for instance, air traffic control radars, is to vary the PRF to identify the correct target range from an ambiguous set [8]. For SAR systems, this scheme has been adapted and introduced as staggered SAR [9]. It, however, not only considerably complicates the system and digital signal processing but also achieves only a limited range ambiguity suppression capability.

Furthermore, it has been proposed to modulate the transmitted signal waveform on a pulse-to-pulse basis to be able to uniquely identify, for a given received echo, the corresponding correct transmit pulse [10], [11]. By doing so, returns from previous transmit pulses can be discerned from the current pulse and the range ambiguity resolved. For instance, using alternating positive and negative chirp rates for a linear frequency modulated (LFM) chirp waveform on successive pulses is discussed in [12]. However, since the waveforms are not orthogonal, the energy from the ambiguous ranges will not be eliminated but rather spread out, resulting in a raised noise floor with the potential to mask out and losing legitimate targets. More elaborate phase coding schemes set the PRF sufficiently high and attempt to separate the ambiguous and unambiguous frequency contributions, i.e., to shift the ambiguity Doppler band outside of the processed Doppler bandwidth [13], [14]. A large PRF, however, is counterproductive as it restricts inevitably the swath width. Another approach tries to generate orthogonal signal waveforms through cyclic frequency shifting requiring rapid frequency hops that may create challenges for hardware implementation [15]. Recently, even more intricate efforts that try to combine the up–down chirp approach with postprocessing algorithms based on dual-focusing [16], [17] or on phase coding schemes [18], [19] have been reported. For future generation SARs, technologically advanced techniques have been suggested, e.g., in [20] and [21] using a phased array antenna with multiple phase centers in elevation to form antenna pattern nulls in the directions of the range-ambiguous returns.

All these techniques have in common that they try to generate clean imagery by eliminating or at least suppressing the range ambiguities with more or less complex technological modifications. Therefore, they cannot readily be applied to improve existing in orbit systems.

An attempt to identify a special case of range ambiguities called grating lobe ambiguities on RCM that are caused by elevation steering toward high incidence angles has been reported in [7]. At large incidence angles ($>50°$), commonly used for vessel detection, the antenna main lobe is steered so far away from physical boresight that a significant grating lobe with similar gain arises opposite the nadir. This grating lobe can superimpose strong echoes from areas more than 1000 km away. The presented algorithm uses repeat-pass imagery to automatically identify these range ambiguities based on the fact that they should always appear exactly at the same location. A similar approach has been reported for general range ambiguities in [22].

B. Azimuth Ambiguities

Azimuth ambiguities arise because the slow-time phase history is strictly not a Doppler band-limited signal based on the "sinc-like" shape of the antenna pattern. Consequently, for a finite PRF, the outer parts of the undersampled signal are aliased and fold back into the processed Doppler bandwidth. These Doppler contributions create additional weaker "ghost" images on either side of the correct image, i.e., are displaced in azimuth and range. Their amplitudes are proportional to the diminishing shape of the antenna pattern outside the mainlobe with respect to the PRF. They are smeared out due to an incorrect mapping of the back-folded frequency components to the processed frequencies [23], [24], [25]. Depending on the chosen PRF, these ambiguous peaks can be strong enough to be detected by the CFAR ship detector, especially for wide-area ScanSAR applications that achieve larger swaths in exchange for reduced PRFs. Azimuth ambiguities arise equally for stationary and moving objects. Since the distance from the ambiguity to the correct scatterer location is typically in the order of several kilometers, they can create false alarms for land scatterers near the transition area between land and water, i.e., close to shore. This problem may be overcome by enlarging the landmask a certain distance into the water. This, however, prevents successful ship detection in littoral zones.

Although the true image for a moving vessel appears displaced through a radial velocity component, the azimuthal distance of the ambiguities from the correct image does not change, i.e., is independent of the vessel's velocity [23]. Since this distance is known (i.e., only depends on SAR system parameters), the identification of vessel ambiguities becomes a deterministic association problem. Albeit tedious, it can be fully automated [22] but can pose problems for many closely spaced ships like, for instance, large commercial fishing fleets or offshore anchored oil tankers acting as floating storage tanks. In [26], it was proposed to detect the vessels by a deep learning approach using convolution

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neural networks (CNNs) instead of a CFAR detector while the ambiguities are identified based on "visual similarity" at the a priori known ambiguity location. Unfortunately, ambiguities must not necessarily resemble the main image due to the aspect angle dependency of the radar cross section (RCS) [27].

Other approaches to suppress azimuth ambiguities are based on dedicated postprocessing techniques. For instance, adaptive spectrum filtering has been proposed to decrease frequency bands containing the aliased contribution based on the shape of the antenna pattern [28], [29], [30]. These require an estimate of the local azimuth-ambiguity-to-signal ratio (AASR), e.g., [31], which, however, needs the ambiguity to appear in the focused SAR image, a fact not always given. In addition, for low PRFs the ambiguity frequency band may cover large parts of the unambiguous (central) Doppler spectrum which would inevitably impact also the useful target spectrum and lead to image distortions and a lower resolution. Even more problematic, the Doppler shift into the higher frequency components caused by a radial velocity of a moving vessel could lead to suppression of desired target energy and hence defeat the purpose. One dedicated example for ship detection tries to filter out the azimuth ambiguities on SAR stripmap images as a preliminary step for an adaptive-threshold cell-averaging CFAR ship detection algorithm [32].

Another proposed technique for azimuth ambiguity detection is based on two range look processing for high-resolution (large bandwidth) SAR so that the individual azimuth ambiguities are misaligned but not the true target [33]. In [34], it is proposed to filter out only the range-dependent spectral signatures created by the wrongly applied range cell migration correction (RCMC) thereby enabling to suppress only the ambiguous energy. Both the techniques are not applicable to coarser resolution space-based ScanSAR beam modes typically used for ship detection.

Similar to the range ambiguity case, technological advanced adaptive beamforming approaches have been proposed for future multiaperture SAR systems and feasibility was recently demonstrated using experimental two-aperture SAR data from TerraSAR-X [35] and RADARSAT-2 [36].

III. Contribution and Organization of this Article

It is interesting to note that to the best knowledge of the authors, all the ambiguity mitigation techniques presented so far have merely been demonstrated on higher resolution imagery in stripmap or spotlight mode and mostly over land to improve feature recognition and image interpretability. There are few publications dealing with improved vessel detection in littoral zones and even fewer for ship detection on open water. For moderate- to low-resolution beam modes using ScanSAR designed specifically for vessel detection, investigations are notably absent. Due to reduced spectral support, their properties are expected to be quite different and their feasibility and ambiguity suppression capabilities are doubtful. In addition, only a very few attempts have been reported to date that try to suppress both range and azimuth ambiguities simultaneously, e.g., [37] an important requirement for vessel detection in coastal areas.

Instead of trying to avoid the emerging of ambiguities from the outset by intricate technological means, or to mitigate already present ones by digital filtering, which always (to a greater or lesser extent) leads to some degradation of the desired information, this article proposes a different approach. It is based on a postprocessing step on the unaltered SAR data and hence does not suffer from image degradations. After CFAR detection and clustering of the individual pixels, each cluster undergoes an additional statistical hypothesis test to check whether the cluster is a true positive or in fact an azimuth or range ambiguity. The test exploits the information contained in a slow-time (azimuth) phase history data chip of the detected cluster around its center location. The test statistic is formulated as a solution of a novel maximum likelihood quotient test (MLQT) for a test problem that assesses the compression (or focusing) gain/loss in azimuth. If the assumed theoretical phase history of a cluster under test does not match that of the measured signal during azimuth refocusing, the discrepancy leads to a reduced depth-of-focus, resulting in blurring of the cluster image. In contrast, when the theoretical azimuth compression filter is matched to the signal, the focusing improves. This discrepancy is evaluated by the MLQT to distinguish between vessel and ambiguity.

This article is organized as follows. After briefly discussing the physical reason for the arising of ambiguities in both the directions, range and azimuth, most important published technologies to mitigate their effects are presented in Section II. The following two sections introduce the proposed statistical hypothesis test and its functionality for range as well as azimuth ambiguity indication, respectively. The novel MLQT statistic and its distributions are derived in the Appendix, which are subsequently used to compute the receiver operating characteristics (ROC) as theoretical figure of merit. The new detectors are then demonstrated and assessed based on extreme cases of real-world data from RCM. This article concludes with a summery and an outlook into potential future work.

IV. Range Ambiguity Indication

After range-compression and stationary RCMC but before azimuth compression (phase history data), the baseband echo of a point scatterer along slow-time t (azimuth) can be expressed to

$$s(t) = a e^{-j2\beta\left(R_b + \frac{v_{\text{rel}}^2}{2R_b}(t-t_b)^2\right)} D[u(t-t_b)] \cdot \text{comb}\left(t - t_b, T_{\text{bcp}}, T_b\right) \quad (1)$$

with a constant amplitude $a \in \mathbb{C}$ and $D[u(t)]$ denoting the two-way antenna pattern. Broadside time t_b and minimum slant range R_b describe the scatterer location. In Earth-centered-Earth-fixed (ECEF) coordinate system, the relative antenna velocity for a stationary scatterer reads $v_{\text{rel}}^2 = v_{\text{sat}} v_g$ in which v_{sat} denotes the satellite's velocity in flight direction

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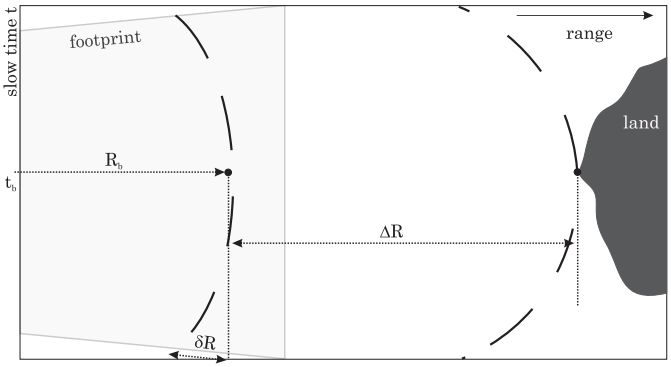


Fig. 1. Schematics of range ambiguity creation.

and v_g the ground speed of the spacecraft. The directional cosine angle history $u(t)$ between the SAR and a scatterer can be approximated by

$$u(t-t_b) = -\frac{v_g}{R_b}(t-t_b) + u_0 \tag{2}$$

with u_0 denoting a possible antenna squint angle that is assumed constant over the acquisition time. The comb-function

$$\mathrm{comb}(t, T_s, T_w) = \sum_{l=-\infty}^{\infty} \mathrm{rect}\left(\frac{t - l\,T_s}{T_w}\right) \tag{3}$$

with

$$\mathrm{rect}\left(\frac{t}{T}\right) = \begin{cases} 0, & t < -\dfrac{T}{2} \\ 1, & -\dfrac{T}{2} \le t \le \dfrac{T}{2} \\ 0, & t > \dfrac{T}{2} \end{cases}$$

mathematically represents the ScanSAR operation of the SAR when full aperture processing is used, i.e., the gaps between the individual bursts of duration T_b within one subbeam are filled with zeros, consistent with a burst cycle period $T_{\mathrm{bcp}} = T_g + T_b$. Although the gap duration T_g and burst duration are chosen to guarantee a desired number of looks $N_{\mathrm{looks}} = T_{\mathrm{sa}}/T_{\mathrm{bcp}}$, obviously the exact location of the individual bursts within the synthetic aperture time T_{sa} for each scatterer is unknown causing the typical ScanSAR scalloping effect. For details on full aperture processing, see [38].

Fig. 1 illustrates the phase history trajectory of a nonmoving scatterer located at $(t = t_b, R = R_b)$ within the footprint of the SAR antenna including its range-dependent range migration

$$\delta R = \sqrt{R^2 + \left(\frac{R\,\varphi_{3\,\mathrm{dB}}}{2}\right)^2} - R \cong \frac{R\lambda^2}{8L^2} \tag{4}$$

where $\varphi_{3\,\mathrm{dB}}$ denotes the azimuth 3-dB beamwidth and L the aperture length of the SAR antenna. The intermittent line depicts the burst nature of the ScanSAR operation when full aperture processing is applied. Another scattering echo for a previous transmit pulse located farther out at $R = R_b + \Delta R$ in slant range with $\Delta R = (c/2\mathrm{PRF})$, potentially originating on land inside or outside the radar footprint, would fall inside the imaged swath and if strong enough create an ambiguity image at $(t = t_b, R = R_b)$. For exemplary RCM system parameters in its ship detection mode listed in Table I, see also Table IV (scene 1), one can note that the increase in range migration between R_b and $R_b + \Delta R$, i.e., $\delta R = (\Delta R\lambda^2/8L^2) = 8.42$ m, may not be negligible but small compared with typical range resolution (see also Table V). In contrast, the phase change is crucial because it causes a defocussing when convolved with the mismatched reference phase in (1). The loss of compression gain (in addition to a resolution loss) lends itself to applying a test of the form

TABLE I
C-BAND ($\lambda = 0.0566$) SAR SYSTEM PARAMETERS

L	PRF	v_{sat}	v_{rel}	R_b	N_g	N_{b}
6.75m	2.764kHz	7.65km/s	7.28km/s	786km	1662	218

$$T = \max\left\{\frac{|(f_1 * z)(t)|^2}{||f_1(t)||^2}\right\} - \max\left\{\frac{|(f_0 * z)(t)|^2}{||f_1(t)||^2}\right\} \tag{5}$$

in which $f_i(t)$ denotes the azimuth compression filters $f_i(t) = s_i^*(-t)$ at $R = R_b \pm i\Delta R$ for $i \in \mathbb{N}$ and $z(t)$ describing the measured phase history signal of the detection at (t_b, R_b) including additive thermal noise. Note that only a far out ambiguity is depicted in Fig. 1 for clarity. However, ambiguities can originate from several previous or later pulses [7]. Nevertheless, analysis has shown that for most operational scenarios, consideration of only the first-order range ambiguities ($i = 1$) is commonly sufficient. Higher order ambiguities, although sometimes visible, tend to remain below the 3MD detection threshold. Obviously, T is measuring the difference in the compression gain, or in other words, the degree of defocussing. If the slow-time signal $z(t)$ occurring at R_b would have in fact stemmed from an ambiguous range, then $T > 0$, while the opposite is true if the signal indeed originated from that specific range.

Writing the sampled slow-time signals as vectors $\mathbf{Z}$, the convolutions can be expressed as matrix multiplications $\mathbf{F}_i\mathbf{Z}$, where $\mathbf{F}_i$ are the convolution matrices generated by vectors $\boldsymbol{f}_i$. Then, one can write (5) to

$$T = \frac{\left|\boldsymbol{e}_m'\mathbf{F}_1\mathbf{Z}\right|^2}{\left\|\boldsymbol{e}_m'\mathbf{F}_1\right\|^2} - \frac{\left|\boldsymbol{e}_m'\mathbf{F}_1\mathbf{Z}\right|^2}{\left\|\boldsymbol{e}_m'\mathbf{F}_0\right\|^2} \tag{6}$$

in which $\boldsymbol{e}_m$ is a zero vector with a single one as the mth element, therewith extracting the mth row of matrix $\mathbf{F}_i$. This row vector corresponds to the convolution result at the location of the maximum peak, which is known as it corresponds to the image location of the detection under test.

To analyze the capability of T, it is instructive to consider (6) being the solution of MLQT for the test problem

$$\begin{aligned} H_0 &: \mathbf{Z} = a_0\boldsymbol{f}_{0,m} + \boldsymbol{N} \quad \text{(correct range)} \\ H_1 &: \mathbf{Z} = a_1\boldsymbol{f}_{1,m} + \boldsymbol{N} \quad \text{(ambiguous range)} \end{aligned} \tag{7}$$

where $\boldsymbol{f}_{i,m} = \mathbf{F}_i^H e_m$ are the known signal vectors with unknown complex amplitudes a_i for $i = 0, 1$. The vector

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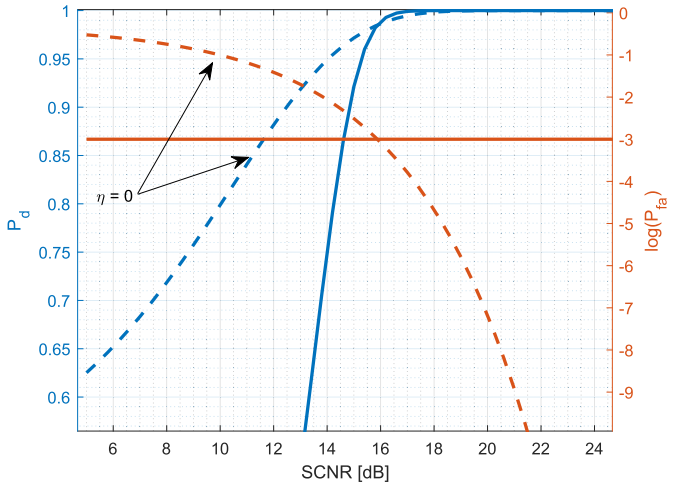


Fig. 2. Probabilities P_d and P_{fa} versus SCNR for the range ambiguity test statistic.

$\boldsymbol{N}$ represents the additive noise contribution modeled as zero mean complex normal distributed with known covariance matrix $\mathbf{R} = \sigma^2\mathbf{I}$. Since the SAR pixels under observation have already been soundly detected by 3MD beforehand, it is reasonable to assume that no clutter suppression is necessary for the subsequent ambiguity test. This means the second test only needs to consider thermal noise and the sea clutter contribution can be omitted. The corresponding test statistic for the problem (7) has been derived in Appendix A to

$$T = \frac{|\boldsymbol{f}_{1,m}^H \mathbf{Z}|^2}{\boldsymbol{f}_{1,m}^H \boldsymbol{f}_{1,m}} - \frac{|\boldsymbol{f}_{0,m}^H \mathbf{Z}|^2}{\boldsymbol{f}_{0,m}^H \boldsymbol{f}_{0,m}} \underset{H_0}{\overset{H_1}{\gtrless}} \eta \tag{8}$$

where η denotes the detection threshold. Note that in Appendix A, the test is derived for the more general case of an additional sea clutter contribution. This means if the interference is not white, one would need to include $\mathbf{R}^{-1}$ into the test, see (19), thereby prewhitening the data vector.

Using the derived distributions of the test statistics under the hypothesis and the alternative in Appendix B, the ROC can be computed as performance metric. Note that for simplicity in the following, without loss of generality, the performance of test (8) is only analyzed for a point scatterer model. In fact, in the Appendices the statistics of an accumulation of $K > 1$ statistically independent scatterers are considered and derived. These results can be used to analyze extended targets, i.e., detected clusters containing more than a single pixel.

Fig. 2 shows the probability of detection P_d and the probability of false alarm P_{fa} versus signal-to-clutter plus noise ratio (SCNR) after processing using the system parameters listed in Table I and a CNR = 10 dB. The solid blue curve shows P_d for a chosen CFAR of 10^{-3} for which the detection threshold has been computed via numerical inversion of (30).

The chosen CFAR value demands that under ideal conditions, on average, only one out of a thousand a priori detections shall be falsely indicated as range ambiguity (H_1) although it originated from the correct range (H_0). In other words, one attempts to keep the chance small to lose a potential legitimate vessel. On the other hand, the probability to miss a range-ambiguity (decision for H_0 although H_1 is correct) is given by $P_{\text{miss}} = 1 - P_d$. P_d can be considered a "clean-up success rate" as it represents the number of range ambiguities that are correctly identified as such. One can see from Fig. 2 that 75% of range ambiguities are correctly recognized when SCNR = 14 dB and goes rapidly to 100% if the power levels of the detections are higher. The dashed blue curve shows P_d if the detection threshold is set constant to zero, which means a willingness to surrender the CFAR property for the benefit of simplicity to construct the test (no threshold needs to be computed). The variation in the corresponding P_{fa} is plotted in red (dashed) with respect to the right axis. It can be observed that P_{fa} worsens to $\cong 10^{-2}$ for a detection with SCNR = 14 dB, whereas P_d goes up to about 0.95. In other words, if one is willing to accept one missed vessel out of a hundred detected clusters, one can theoretically expect to correctly identify 95% of all the range ambiguities when the threshold is set to zero.

V. Azimuth Ambiguity Indication

Omitting for simplicity at the moment the comb-function and using the mathematical concept of stationary phase, the Fourier transform of (1) can be written in closed form to

$$\begin{aligned} S(f) &= \sum_m S_m(f) \\ &= c \sum_m e^{j2\pi(f+m\text{PRF})\left(t_b + \frac{v_{\text{sr}}}{v_{\text{rel}}^2} R_b\right)} D\left[u(f+m\text{PRF})\right] \\ &\quad \cdot e^{-j\frac{\pi\lambda R_b}{2v_{\text{rel}}^2}(f+m\text{PRF})^2} \end{aligned} \tag{9}$$

for $m \in \mathbb{Z}$, with direction Doppler dependency

$$u(f) = -\frac{\lambda}{2v_{\text{sat}}} f + \frac{v_{\text{sr}}}{v_{\text{sat}}} + u_0 \tag{10}$$

and some constant $c \in \mathbb{C}$. Considering a common shape for the antenna pattern, e.g., sinc-like, in most cases, similar to the range-ambiguity problem, it is sufficient to restrict $|m| \leq 1$ without loss of generality. A potential slant range velocity component is denoted as v_{sr}. Analogous to the range ambiguities in Section IV, the intention is to develop a test for azimuth ambiguities based on (5) and (6).

The general idea behind the functionality of the test is illustrated in Fig. 3 which depicts the process of azimuth compression in the Doppler slow-time (f, t) domain, where the received signal at a given range R_b possesses the linear relationship

$$f(t) = -\frac{2v_{\text{rel}}^2}{\lambda R_b}(t - t_b) - \frac{2v_{\text{sr}}}{\lambda}. \tag{11}$$

Note that the radial velocity component v_{sr} is omitted from here onward since it simply causes the identical azimuthal displacement for both, the moving vessel's main image and its

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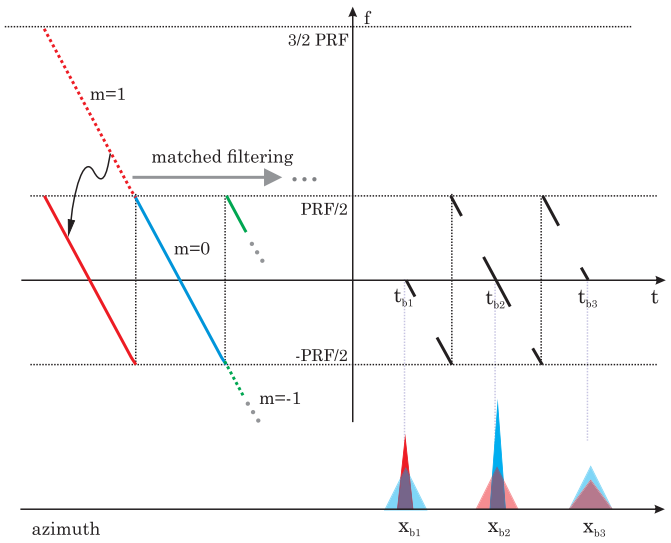


Fig. 3. Schematics of the azimuth ambiguity indication test.

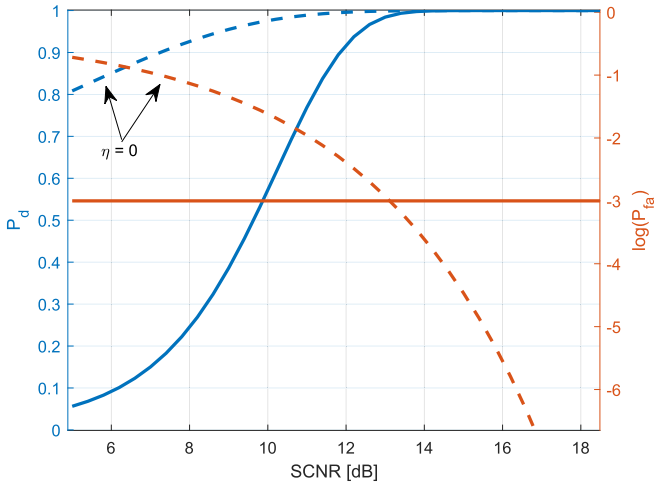


Fig. 4. Probabilities P_d and P_{fa} versus SCNR for the azimuth ambiguity test statistic.

ambiguities without impacting their resolutions [36]. In addition, an along-track velocity component can also be neglected for medium- and low-resolution space-based SAR. It has been well-established that smearing of the impulse response (vessel image) for space-based geometries by an along-track velocity component is very small and virtually negligible. This is mainly based on the fact that ground moving target velocities (including vehicles on land) are much smaller than the spacecraft velocity of about 7500 m/s, which means the Doppler slope only changes insignificantly. That is also exactly the reason why it is extremely difficult to estimate the along-track component of the target's velocity from space in contrast to airborne SAR [36]. Range accelerations stemming from heaving and swaying in the waves, on the other hand, can potentially affect the appearance of the vessel image but it will do so in the same way for both, the correct image and the corresponding azimuth ambiguities.

On the right-hand side in black, the somewhat undersampled burst-like receive signal of one detection is depicted leading to wraparounds of some parts of its spectrum at $f = \pm\text{PRF}/2$. On the left-hand side, the reference signal (azimuth compression filter) with identical slope is plotted for two Doppler bands $m = 0, 1$ ($m = -1$ omitted for clarity). Obviously, the spectral components for $m = 1$ (red dashed) wrap also into the unambiguous band (red solid). The convolution process can be illustrated as moving the filter function along positive slow-time and integrating the energy along the trajectory when two lines coincide. Underneath the (f, t)-plane the image domain is depicted showing the compression results along azimuth. The filtering of a target's unambiguous frequency band with the blue filter ($m = 0$) generates a sharp peak at location $x_{b_2} = v_g t_{b_2}$ as the entire spectral support (PRF) is integrated and the phases match. The compression result of the target's unambiguous frequency band with the red filter ($m = 1$), in contrast, is considerably smaller due to phase mismatch. The impulse response for the blue filter with the upper ambiguous spectral component of the target at x_{b_1} is much smaller because of the reduced spectral support and a phase mismatch. In contrast, when the same spectral part of the target is compressed with the red filter, its response is larger since the phases match even if the spectral support is reduced.

In summary, it can be noted that by comparing the peaks for each Doppler band filter, qualitatively $T < 0$ for the unambiguous part of the potential target signal, whereas $T > 0$ for the ambiguous part, which can be exploited to distinguish between a true SAR detection and its ambiguous counterpart. Mathematically, one can use the same test statistic, (8), with the difference that here the signal vectors $\boldsymbol{f}_i$ are the Fourier back-transformed signal contributions for each Doppler band in (9), i.e.,

$$\boldsymbol{f}_i = \mathfrak{F}^{-1}\left\{\left[S_i(f_0), \ldots, S_i(f_N)\right]'\right\} \tag{12}$$

for $i = -1, 0, 1$ assuming N Doppler frequency samples. As for the range ambiguity test, the ROC of the azimuth ambiguity test for the parameters in Table I and a CFAR of 10^{-3} are plotted as solid curves in Fig. 4. Compared with Fig. 2, a performance improvement can be observed since a detection with mere 12-dB SCNR is correctly identified as azimuth ambiguity nearly 90% of the time. Again, assuming as a rule of thumb that a minimum of 14-dB SCNR is required for 3MD to be able to detect any vessel or ambiguity, P_d of nearly 1 can be achieved for $P_{\text{fa}} = 10^{-3}$. By setting the threshold constant to zero (dashed curves) yields a near-perfect 100% P_d and P_{fa} smaller than 10^{-3} for clusters with SCNR > 14 dB. In summary, Fig. 4 suggests an almost perfect indication of Doppler ambiguities for the chosen RCM system parameter set. Note that the proposed Doppler ambiguity test does not rely on displacement as decisive parameter but on the comparison of compression gains.

VI. Experimental Results

A. Setup

To confirm feasibility and to demonstrate the performance of the proposed ambiguity indicators on real data, three subbeams within two extremely challenging RCM ScanSAR

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TABLE II
RCM DATASETS USED TO EVALUATE THE AMBIGUITY INDICATORS

ID	Mode	Location	Date	Pol	Beam	AIS
1655887	SCSDA	Gibraltar	28/01/2020	HH	2	18
1655887	SCSDA	Gibraltar	28/01/2020	HH	5	23
2075627	SCSDA	Baltic Sea	30/04/2022	HH	10	3

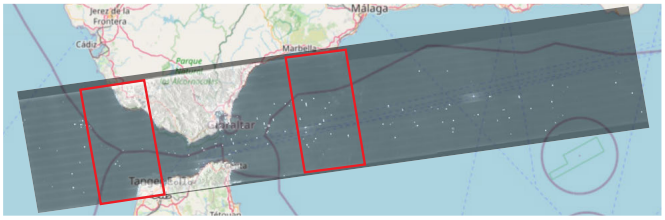


Fig. 5. Footprint of the RCM-2 data take 1 655 887 with highlighted subbeams two and five. Background ©openstreetmap.org.

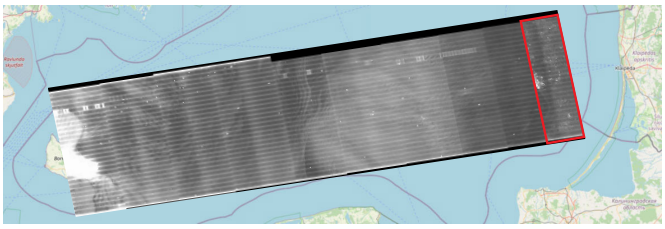


Fig. 6. Footprint of the RCM-1 data take 2 075 627 with analyzed subbeam ten highlighted. Background ©openstreetmap.org.

TABLE III
METEOROLOGICAL DATA DURING DATA TAKES INCLUDING SIGNIFICANT WAVE HEIGHT (SWH)

Scene	ID	SWH [m]	Wind [kts]	sea state
1	1655887	1	8	3
2	1655887	0.5	10	3
3	2075627	0.3	6	2

datasets are analyzed. These include an excessive number of ambiguities in range and azimuth alongside several vessels. Table II summarizes the relevant information. The last column lists the number of present vessels, for which corresponding ground-based automatic identification system (AIS) information was gathered from the Maritime Safety and Security Information System (MSSIS) database [39]. Figs. 5 and 6 show the geographic locations of the swaths including the analyzed subbeams highlighted by red squares. While the data take over Gibraltar is an example for ambiguities originating from scatterers within the imaged scene, the Baltic Sea data are a striking example of many present range ambiguities on the open ocean originating from an area outside the footprint, namely, the Baltic State of Lithuania to the East.

Table III lists the weather conditions during the time of data recording, and it can be observed that the sea state for all the scenes is only 3 or lower. This leads to believe that almost all the false alarms within the imagery are most likely caused by ambiguities and not by breaking waves (sea spikes). Note that the dedicated ScanSAR ship detection beam mode (SCSDA) on RCM is not an image mode per se but optimized for the detectability of vessels similar to the Maritime Satellite Surveillance Radar (MSSR) ScanSAR modes on RADARSAT-2 [40]. Therefore, each subbeam typically uses a different set of SAR parameters, i.e., chirp bandwidth B, PRF, sampling frequency f_s, number of samples within one burst, and gap N_b, N_g. Together with geometric parameters R_b (subbeam center) and v_{sat} they determine the slant range sampling width δ_{sr} and resolution Δ_{sr} as well as the counterparts in azimuth δ_{az} and Δ_{az}, respectively. The corresponding values are listed in Tables IV and V.

TABLE IV
RELEVANT SAR AND GEOMETRIC PARAMETERS

f_s	PRF	B	v_{sat}	R_b	N_{b}	N_{g}
86MHz	2.674kHz	76MHz	7.65km/s	768.5km	218	1662
54MHz	2.835kHz	50MHz	7.65km/s	812.6km	154	1774
27MHz	2.432kHz	25MHz	7.64km/s	1017km	144	1523

TABLE V
SAMPLE SPACINGS δ AND RESOLUTIONS Δ

Scene	δ_{sr}	δ_{az}	Δ_{sr}	$\Delta_{\mathrm{az,burst}}$
1	1.74 m	2.77 m	1.97 m	26.03 m
2	2.79 m	2.70 m	3 m	47.52 m
3	5.58 m	3.14 m	6 m	59.15 m

The experimental setup was as follows. After full aperture processing of the phase history data using the chirp scaling processor [41], the 3MD sea clutter model-based detector was applied [6] with a CFAR of 10^{-10} [3]. Although no multilooking was applied, the number of looks n was treated as an unknown parameter and estimated alongside the sea clutter parameters to improve the quality of the statistical fit. Where required, a manually created landmask was applied that was deliberately chosen to be as close as possible to shore to be able to analyze azimuth land ambiguities.[1] The sea clutter parameters for each tile are estimated using the approach proposed in [42], and the tile size was chosen to 1 km^2. The entire estimation for each tile only takes a fraction of a second in MATLAB© on a conventional workstation. Subsequently, a standard clustering algorithm was applied to the individually detected pixels. It is based on a given maximum distance to group closely located pixels into clusters representing individual target detections. It is important to note that no further selections were applied, for instance, rejecting single-pixel clusters or removing clusters that do not appear "shipness"-like [1].

Afterward, for each detected pixel cluster, the azimuth ambiguity test was performed for $m = \pm 1$ using as threshold a constant $\eta = 0$, and if successful, i.e., an azimuth ambiguity declared, a red |-marker plotted onto the cluster center and the algorithm terminated. If the test came out negative, the range ambiguity hypothesis test was performed for $i = \pm 1$, and if successful a yellow $-$marker was used and the algorithm

[1]Note that operationally applied landmasks commonly use a buffer zone to mask out sidelobes and azimuth ambiguities of strong land scatterers.

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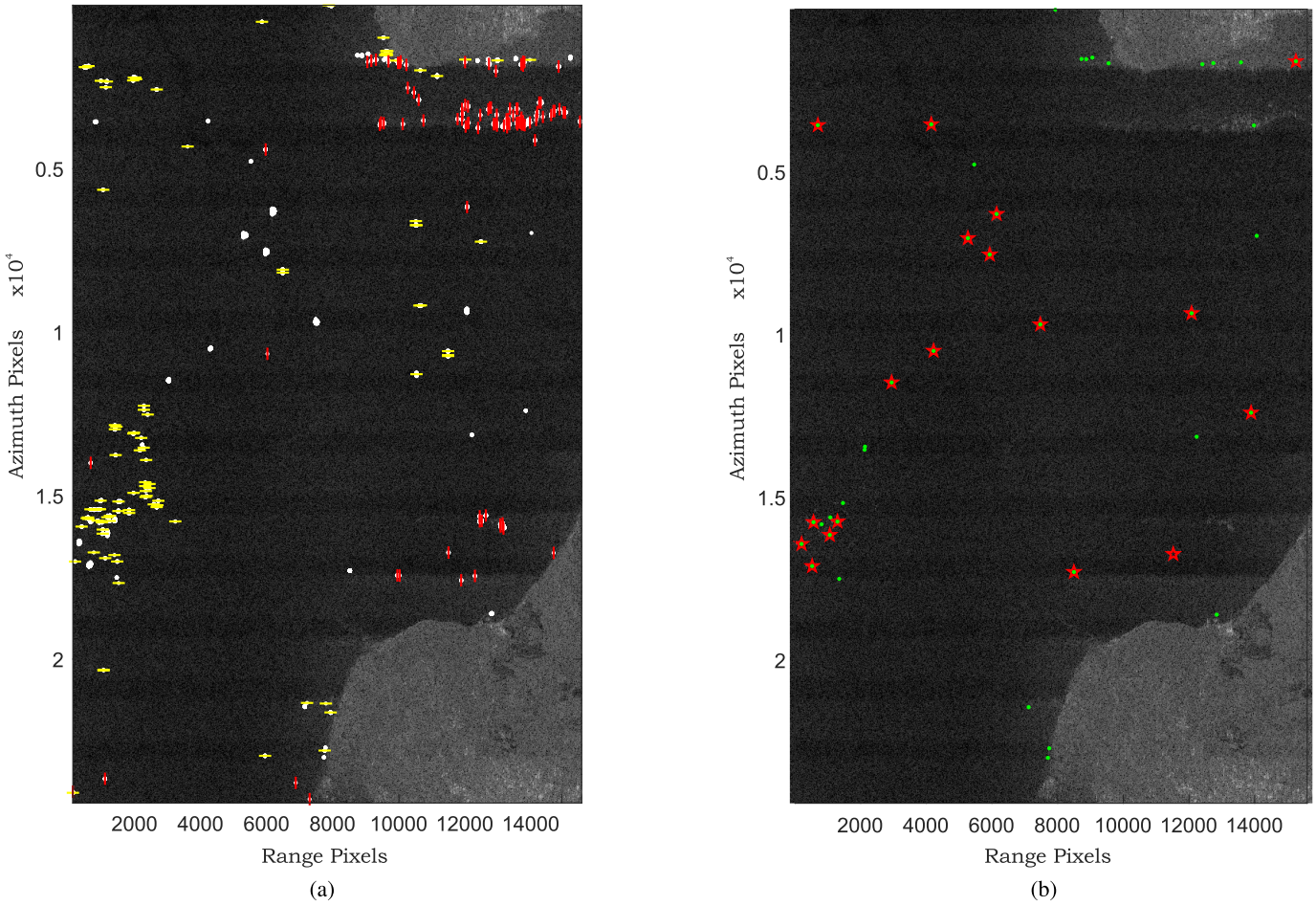


Fig. 7. Ambiguity indication results for subbeam two of the Gibraltar scene. RCM Imagery © Government of Canada (2020). RADARSAT is an official mark of the Canadian Space Agency. (a) 242 3MD-detected clusters (white dots), range ambiguities (yellow -), and azimuth ambiguities (red |). (b) 18 AIS-confirmed vessels (red stars) and 21 remaining false positives (green dots).

stopped. Only if both the tests came out negative, a positive true detection was declared for the cluster and a green dot assigned. Note the entire process from 3MD detection, over clustering to ambiguity testing ran autonomously without user interaction. Note that there might be a small chance that a detection is identified as both, a range and an azimuth ambiguity simultaneously, or it may be falsely detected as a range ambiguity although the opposite was true (azimuth ambiguity) and vice versa. However, the way the test is set up sequentially, it will stop when either one criterion is triggered and hence does not pose a problem because at the end the objective is to clean up the 3MD detection map without losing any true vessels.

B. Results

Fig. 7(a) shows the SAR intensity image of subbeam two of scene one (Gibraltar) including all 242 detected clusters plotted as white dots. The scene is insofar interesting as it combines many azimuth ambiguities below and above the two continents and many range ambiguities alongside 18 AIS-confirmed vessels. Since the first ambiguous range equals $(c/2\text{PRF}) = 54$ km, most range ambiguities are stemming from strong land scatterers to the east in subbeam three (see Fig. 5). As mentioned before, the results of the individual ambiguity tests are superimposed with red (azimuth) and yellow (range) markers. In conjunction with Fig. 7(b), it can be observed that virtually all the azimuth land ambiguities near the shore are correctly identified as well as the majority of the range ambiguities on the open water. From the 18 AIS-confirmed vessels, 17 are correctly indicated as true positives. The remaining vessel at pixel location (11.5k, 16.5k) is a small fishing vessel of length 16 m and was not detected by 3MD. From the original 242-17 = 225 false positives, only 21 remain after the ambiguity indication tests, which corresponds to a clean-up rate (CUR) of 90.7%. The theoretical analysis in Fig. 2 predicted a CUR for range ambiguities in the order of 95%. It is noteworthy that about ten of the remaining true positives are extremely close to the land (bottom and top right) and are likely caused by strong sidelobes of land scatterer near the shoreline. For such sidelobes, the indication as not being an ambiguity is actually correct, which brings the CUR for this particular scene closer to the expected one.[2]

[2]Commonly, the landmask buffer would typically somewhat extended to mask out these sidelobe contributions.

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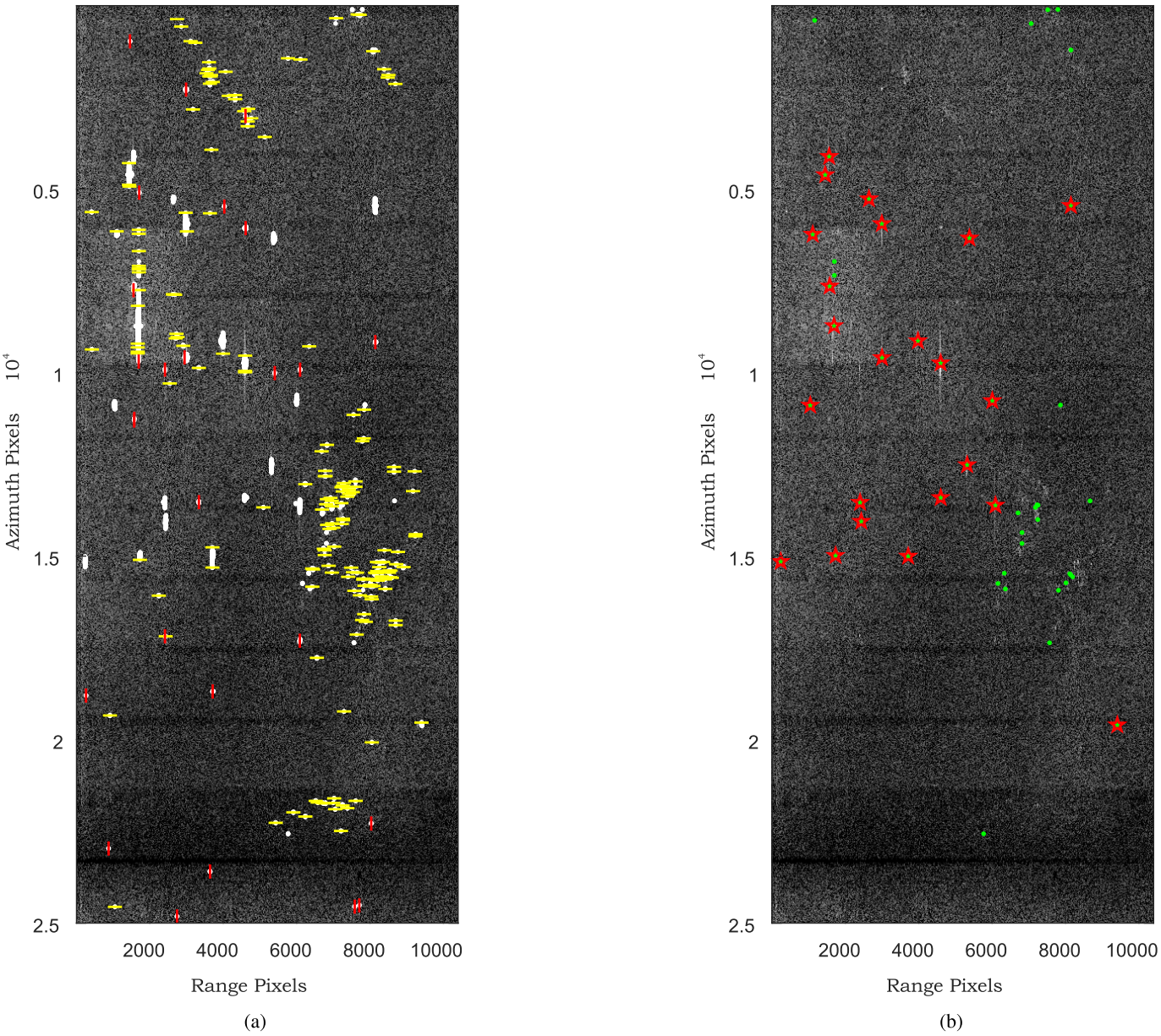


Fig. 8. Ambiguity indication results for subbeam five of the Gibraltar scene. RCM Imagery © Government of Canada (2020). RADARSAT is an official mark of the Canadian Space Agency. (a) 267 3MD-detected clusters (white dots), range ambiguities (yellow -), and azimuth ambiguities (red |). (b) 23 AIS-confirmed vessels (red stars) and 25 remaining false positives (green dots).

The fifth subbeam of the Gibraltar scene in Fig. 8(a) shows 267 3MD-detected clusters alongside 23 AIS-confirmed vessels. The scene is characterized by many strong range scatterers originating from land to the west at a distance of 52.9 km. Although the subbeam does not contain land, it is interesting to note several azimuth ambiguities mostly in the center left. These are in fact Doppler ambiguities of vessels. Fig. 8(b) shows that all 23 vessels have been correctly indicated as true positives and a remaining number of 25 false positives that are most likely range ambiguities. At closer look, it turns out that three of these can be associated with an extremely strong smeared out vessel in the upper left and therefore could be avoided by a more sophisticated clustering algorithm. Considering this, the CUR for scene two becomes 90.6%.

Fig. 6 illustrates the swath width of the Baltic Sea scene confirming an open water data take. However, many strong scatterers from the Lithuanian coast 61.6 km to the east transition into the imagery as range ambiguities. Fig. 9(a) substantiates this assumption and shows a total of 455 detection clusters although only three vessels are confirmed by AIS to be in the image. Running the proposed ambiguity tests confirms that 411 of these clusters are either an azimuth or a range ambiguity while all three vessels are indicated as true positives in Fig. 9(b). Hence, the CUR for this subbeam is 90.9%. Note that the number of samples per burst for the last two scenes

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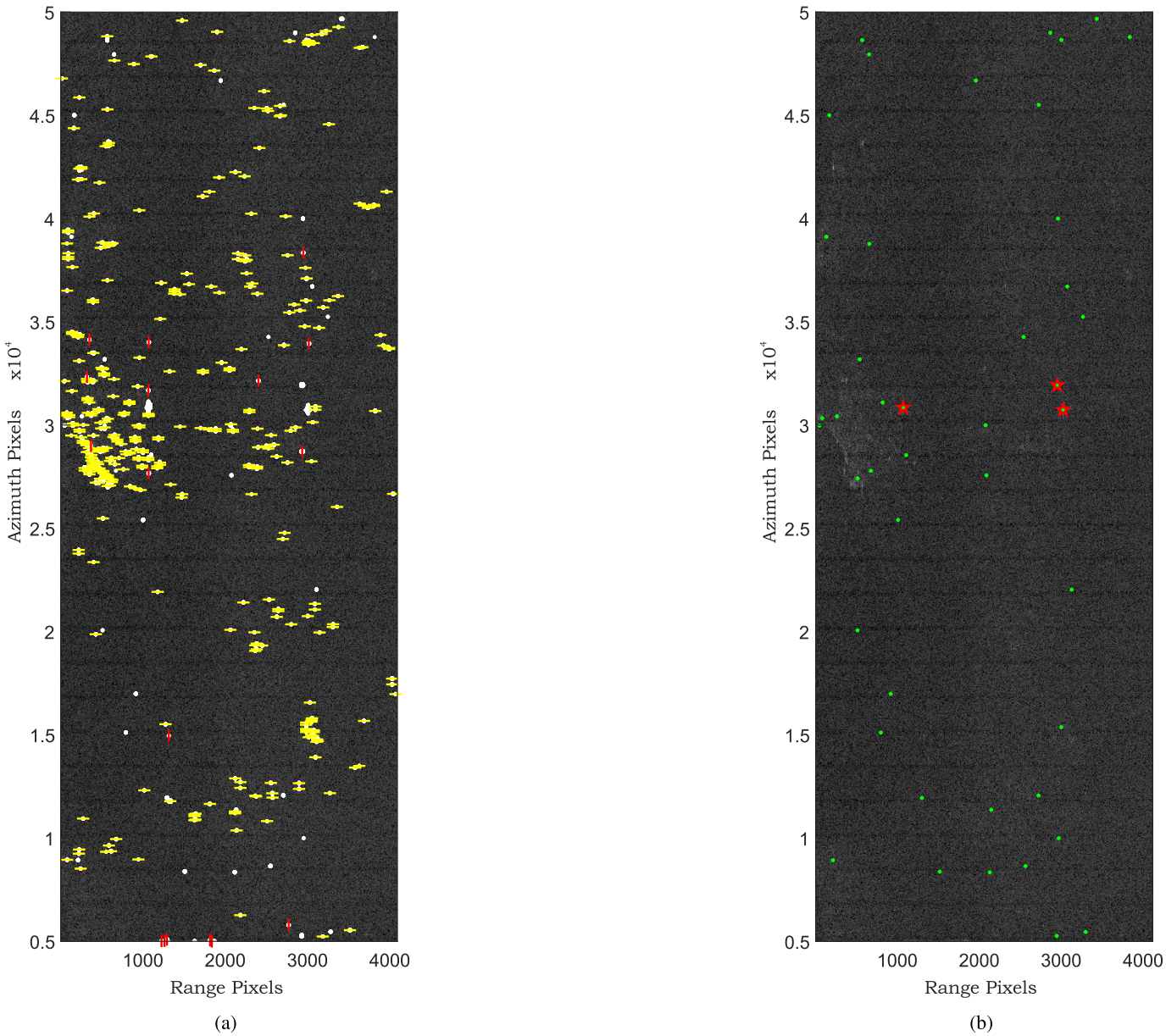


Fig. 9. Ambiguity indication results for subbeam ten of the Baltic Sea scene. RCM Imagery © Government of Canada (2022). RADARSAT is an official mark of the Canadian Space Agency. (a) 455 3MD detected clusters (white dots), range ambiguities (yellow -), and azimuth ambiguities (red |). (b) Three AIS-confirmed vessels (red stars) and 41 remaining false positives (green dots).

is even further reduced by about 30% compared with scene one (see Table IV), and therefore, a somewhat reduced CUR is expected.

VII. Summary and Conclusions

Vessel detection ScanSAR beam modes on modern SAR spacecraft can show, on one hand, an exceedingly large number of range ambiguities, which is first and foremost a result of the required large incidence angles. On the other hand, this demands lower PRFs, which in turn amplifies the creation of azimuth ambiguities originating from scatterer at land and sea. This article developed a new statistical hypothesis test enabling to discriminate whether or not a detected bright pixel cluster is a true positive (vessel) or a false positive/alarm (ambiguity). The test works automatically without manual interaction. It does not require any elaborate spacecraft modification (multiple receive channel, adaptive beamforming, phase codes, etc.) or digital signal processing techniques that try to suppress or mitigate the ambiguous returns prior to ship detection. Therefore, the image quality is not degraded and the only prerequisite of this technique is to access the phase history SAR data, making it immediately applicable to the existing SAR systems and recorded datasets.

The basic principle of the test is based on the comparison of the focusing depth (compression gain) when the matched filter is adapted to either the true expected target model or the ambiguous return. Since it works on a small data chip around the detected cluster, it is computationally inexpensive and fast. The statistical distribution of the test has been theoretically derived for the hypothesis and the alternative, thereby enabling the computation of performance metrics such as ROC. The ROC generated for typical RCM system parameters led to

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expect CURs above 90% for range ambiguities and near 100% for azimuth ambiguities, respectively. These theoretical values were assessed and confirmed using three very challenging RCM datasets. Azimuth ambiguity indication was shown to work successfully for both, stationary and moving scatterer such as vessels. Since azimuth ambiguities are correctly indicated with very high accuracies, it permits ship detection closer to shore than performed today in most operational settings, i.e., landmasks must not protrude into the water for hundreds of meters or even kilometers. In summary, a majority of false ambiguous alarms can be discarded thereby mitigating the burden on analysts and improving operational utility considerably.

Finally, it is important to mention that the presented theoretical performance analysis was done for ScanSAR modes with a large number of subbeams, i.e., a significantly reduced azimuth resolution. The ROCs for higher resolution modes using larger parts of the Doppler spectrum, through fewer subbeams (either larger number of samples per burst or stripmap SAR with full spectral support), are predicted to perform with even higher CURs near 100% in azimuth and range (not shown here).

Appendix A
Maximum Likelihood Quotient Test (MLQT)

This section develops the MLQT based on a general statistical model that covers both the cases, range and azimuth ambiguities. The decision whether the signal corresponds to a true detection or an ambiguity shall be based on a statistical test in a Neyman–Pearson sense that discriminates between the hypothesis H_0 and an alternative H_1

$$H_0:\ \boldsymbol{Z}_k = a_{0,k}\,\boldsymbol{S}(\theta_0) + \boldsymbol{C}_k + \boldsymbol{N}_k \quad \text{(true detection)} \tag{13}$$

$$H_1:\ \boldsymbol{Z}_k = a_{1,k}\,\boldsymbol{S}(\theta_1) + \boldsymbol{C}_k + \boldsymbol{N}_k \quad \text{(ambiguity)} \tag{14}$$

based on N complex measurements $Z_{1,k}, \ldots, Z_{N,k}$, concatenated into vectors $\boldsymbol{Z}_k$ for $k = 1, \ldots, K$. K denotes the number of statistical independent sample vectors, for instance, related to K pixels in an extended detection cluster. From hereon, vectors are written in slanted and bold font while matrices use straight bold letters. The vector $\boldsymbol{S}$ comprises N signal samples along slow-time (azimuth) in the range-compressed, azimuth-uncompressed data domain also called phase history data. All the complex amplitudes $a_{i,k} \in \mathbb{C}$ for $i \in \{0, 1\}$ are assumed unknown, whereas the signal parameters $\theta_i \in \mathbb{R}$ are considered known. The additive sea clutter plus noise contributions $\boldsymbol{W}_k = \boldsymbol{C}_k + \boldsymbol{N}_k$ are modeled as identically independent complex normal distributed with zero mean and known covariance matrix $\mathbf{R} \in \mathbb{C}^{n\times n}$, i.e., $\boldsymbol{W}_k \sim \mathcal{N}_n^{\mathbb{C}}(\boldsymbol{0}, \mathbf{R})$ for $k = 1, \ldots, K$. For notational convenience, let the signal vectors be written as $\boldsymbol{S}_i := \boldsymbol{S}(\theta_i)$ for $i \in \{0, 1\}$ and ^{H} denote complex conjugate transpose. Using the multivariate complex normal pdf

$$f_{\boldsymbol{Z}_1,\ldots,\boldsymbol{Z}_k}(\boldsymbol{Z}_1, \ldots, \boldsymbol{Z}_k) = \frac{1}{\pi^N \det(\mathbf{R})} \times \prod_{k=1}^{K} \exp\!\left(-(\boldsymbol{Z}_k - a_{i,k}\boldsymbol{S}_i)^H \mathbf{R}^{-1} (\boldsymbol{Z}_k - a_{i,k}\boldsymbol{S}_i)\right) \tag{15}$$

in which $\det(\cdot)$ means the determinant, the logarithm of the maximum-likelihood ratio L after omission of constant terms becomes

$$\begin{aligned} L &= \sum_{k=1}^{K}\sum_{i=0}^{1} (-1)^i \left(-\left(\boldsymbol{Z}_k - a_{i,k}\boldsymbol{S}_i\right)^H \mathbf{R}^{-1} \left(\boldsymbol{Z}_k - a_{i,k}\boldsymbol{S}_1\right)\right) \\ &= \sum_{k=1}^{K}\sum_{i=0}^{1} (-1)^i \left(a_{i,k}^H \boldsymbol{S}_i^H \mathbf{R}^{-1} \boldsymbol{Z}_k + a_{i,k}\, \boldsymbol{Z}_k^H \mathbf{R}^{-1} \boldsymbol{S}_i \right. \\ &\qquad \left. - |a_{i,k}|^2 \boldsymbol{S}_i^H \mathbf{R}^{-1} \boldsymbol{S}_i - \boldsymbol{Z}_k^H \mathbf{R}^{-1} \boldsymbol{Z}_k\right). \end{aligned} \tag{16}$$

According to [43], the derivative of L in (16) with respect to the complex parameter $a_{i,k}$ yields

$$\frac{\partial L}{\partial a_{i,k}} = 2\boldsymbol{S}_i^H \mathbf{R}^{-1} \boldsymbol{Z}_k - 2a_{i,k}\, \boldsymbol{S}_i^H \mathbf{R}^{-1} \boldsymbol{S}_i. \tag{17}$$

Setting (17) to zero provides unbiased amplitude estimators

$$\hat{a}_{i,k} = \frac{\boldsymbol{S}_i^H \mathbf{R}^{-1} \boldsymbol{Z}_k}{\boldsymbol{S}_i^H \mathbf{R}^{-1} \boldsymbol{S}_i}. \tag{18}$$

Inserting the estimators into (16) results in the MLQT test statistic

$$T = \sum_{k=1}^{K} \frac{|\boldsymbol{S}_1^H \mathbf{R}^{-1} \boldsymbol{Z}_k|^2}{\boldsymbol{S}_1^H \mathbf{R}^{-1} \boldsymbol{S}_1} - \sum_{k=1}^{K} \frac{|\boldsymbol{S}_0^H \mathbf{R}^{-1} \boldsymbol{Z}_k|^2}{\boldsymbol{S}_0^H \mathbf{R}^{-1} \boldsymbol{S}_0} = \sum_{k=1}^{K} T_k \tag{19}$$

where each T_k corresponds to a difference between two correlated Gamma-distributed random variables.

To understand the principle functionality of the test, it is instructive to look at the expected value of one T_k under H_0, i.e., $\boldsymbol{Z}_k \sim \mathcal{N}_n^{\mathbb{C}}(a_0\boldsymbol{S}_0, \mathbf{R})$

$$\begin{aligned} \mathsf{E}\, T_k|_{H_0} &= \frac{\boldsymbol{S}_1^H \mathbf{R}^{-1}\, \mathsf{E}\, \boldsymbol{Z}_k \boldsymbol{Z}_k^H \mathbf{R}^{-1} \boldsymbol{S}_1}{\boldsymbol{S}_1^H \mathbf{R}^{-1} \boldsymbol{S}_1} - \frac{\boldsymbol{S}_0^H \mathbf{R}^{-1}\, \mathsf{E}\, \boldsymbol{Z}_k \boldsymbol{Z}_k^H \mathbf{R}^{-1} \boldsymbol{S}_0}{\boldsymbol{S}_0^H \mathbf{R}^{-1} \boldsymbol{S}_0} \\ &= \frac{|a_0|^2}{\boldsymbol{S}_1^H \mathbf{R}^{-1} \boldsymbol{S}_1} \left(|\boldsymbol{S}_1^H \mathbf{R}^{-1} \boldsymbol{S}_0|^2 - \boldsymbol{S}_1^H \mathbf{R}^{-1} \boldsymbol{S}_1\, \boldsymbol{S}_0^H \mathbf{R}^{-1} \boldsymbol{S}_0\right) \\ &= |a_0|^2 \boldsymbol{S}_0^H \mathbf{R}^{-1} \boldsymbol{S}_0 \left(|\rho|^2 - 1\right) \end{aligned} \tag{20}$$

where E denotes the expectation operator and

$$\rho = \frac{\boldsymbol{S}_1^H \mathbf{R}^{-1} \boldsymbol{S}_0}{\sqrt{\boldsymbol{S}_1^H \mathbf{R}^{-1} \boldsymbol{S}_1}\sqrt{\boldsymbol{S}_0^H \mathbf{R}^{-1} \boldsymbol{S}_0}}. \tag{21}$$

Using the Cauchy–Schwarz inequality, it is easy to see that $|\rho| \le 1$ leading to $\mathsf{E}\, T_k|_{H_0} \le 0$. In the same fashion, one can show $\mathsf{E}\, T_k|_{H_1} \ge 0$. Accordingly, it is expected that the bulk of the pdf under H_0 is shifted into the negative and under H_1 into the positive. Depending on the angular separation of $\boldsymbol{S}_1$ and $\boldsymbol{S}_0$ and the power level $|a_0|^2$, the gap between the pdfs can be smaller or larger, where the latter improves the probability of detection P_d; see Appendix B.

Appendix B
Distribution of MLQT Statistic

Let us assume $K = 1$ for the moment and denote the test statistic under H_0 as $T_0 := T|_{H_0}$ and rewrite as

$$T_0 = |X_1|^2 - |X_2|^2$$

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for the 2-D complex normal random vector

$$\boldsymbol{X} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim \mathcal{N}_2^{\mathbb{C}}(\boldsymbol{\mu}_X, \mathbf{K}_X) \tag{22}$$

with

$$\boldsymbol{\mu}_X = a_0\sqrt{\boldsymbol{S}_0^H\mathbf{R}^{-1}\boldsymbol{S}_0}\begin{bmatrix}\rho \\ 1\end{bmatrix}, \quad \text{and } \mathbf{K}_X = \begin{bmatrix} 1 & \rho \\ \rho^H & 1 \end{bmatrix} \tag{23}$$

where $\rho = (\boldsymbol{S}_1^H\mathbf{R}^{-1}\boldsymbol{S}_0/(\boldsymbol{S}_1^H\mathbf{R}^{-1}\boldsymbol{S}_1)^{1/2}(\boldsymbol{S}_0^H\mathbf{R}^{-1}\boldsymbol{S}_0)^{1/2})$.

Rewriting the difference as the quadratic product $T_0 = \boldsymbol{X}^H\mathbf{D}\boldsymbol{X}$ with $\mathbf{D} = \left[\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right]$ permits to use the result by Janssen and Stoica [44] to derive the expectation

$$\begin{aligned} \mathsf{E}\, T_0 &= \operatorname{tr}\{\mathbf{D}\mathbf{K}_X\} + \boldsymbol{\mu}_X^H\mathbf{D}\boldsymbol{\mu}_X \\ &= |a_0|^2\boldsymbol{S}_0^H\mathbf{R}^{-1}\boldsymbol{S}_0\left(|\rho|^2 - 1\right) \end{aligned} \tag{24}$$

and variance

$$\begin{aligned} \operatorname{Var} T_0 &= \operatorname{tr}\left\{\mathbf{D}\left(\boldsymbol{\mu}_X\boldsymbol{\mu}_X^H + \mathbf{K}_X\right)\mathbf{D}\left(\boldsymbol{\mu}_X\boldsymbol{\mu}_X^H + \mathbf{K}_X\right)\right\} \\ &\quad - \left(\boldsymbol{\mu}_X^H\mathbf{D}\boldsymbol{\mu}_X\right)^2 \\ &= \operatorname{tr}\{\mathbf{D}\mathbf{K}_X\mathbf{D}\mathbf{K}_X\} + 2\boldsymbol{\mu}_X^H\mathbf{D}\mathbf{K}_X\mathbf{D}\boldsymbol{\mu}_X \\ &= 2\left(1 - |\rho|^2\right)\left(1 + |a_0|^2\boldsymbol{S}_0^H\mathbf{R}^{-1}\boldsymbol{S}_0\right). \end{aligned} \tag{25}$$

Furthermore, let us introduce a prewhitening, i.e.,

$$T_0 = \mathbf{U}^H\mathbf{K}_X^{\frac{1}{2}}\mathbf{D}\mathbf{K}_X^{\frac{1}{2}}\mathbf{U} = \mathbf{U}^H\boldsymbol{\Omega}\mathbf{U} \tag{26}$$

with $\boldsymbol{\Omega} = \mathbf{K}_X^{\frac{1}{2}}\mathbf{D}\mathbf{K}_X^{\frac{1}{2}}$ and $\mathbf{U} \sim \mathcal{N}_2^{\mathbb{C}}\left(\mathbf{K}_X^{-\frac{1}{2}}\boldsymbol{\mu}_X, \mathbf{I}_2\right)$ where $\mathbf{I}_2$ denotes the 2-D identity matrix. After some basic algebra, we get

$$\begin{aligned} \mathbf{K}_X^{\frac{1}{2}} &= \begin{bmatrix} \dfrac{\alpha_1}{2} & \dfrac{\rho}{\alpha_1} \\ \dfrac{\rho^*}{\alpha_1} & \dfrac{\alpha_1}{2} \end{bmatrix} \\ \mathbf{K}_X^{-\frac{1}{2}} &= \begin{bmatrix} \alpha_1 & \dfrac{|\rho|}{\rho^*}\alpha_2 \\ \dfrac{|\rho|}{\rho}\alpha_2 & \alpha_1 \end{bmatrix} \end{aligned} \tag{27}$$

with $\alpha_i = (1 - |\rho|)^{1/2} + (-1)^{i-1}(1 + |\rho|)^{1/2}$ for $i = 1, 2$. Using (27), the inner matrix in (26) becomes diagonal $\boldsymbol{\Omega} = \operatorname{diag}(\omega_1, \omega_2)$ with $\omega_i = (-1)^{i-1}(1 - |\rho|^2)^{1/2}$, and the quadratic product reads

$$T_0 = \frac{\sqrt{1 - |\rho|^2}}{2}\left(|U_1|^2 - |U_2|^2\right). \tag{28}$$

Since $\mathbf{U} \sim \mathcal{N}_2^{\mathbb{C}}(\mathbf{K}_X^{-\frac{1}{2}}\boldsymbol{\mu}_X, \mathbf{I}_2)$, (28) represents a scaled difference between two independent noncentral chi-squared random variables with 2 degrees of freedom each. No closed-form solution of the pdf for the difference is known. However, the pdf of (28) can be accurately evaluated numerically through the infinite integral

$$f_{T_0}(t_0) = \frac{2}{\sqrt{1 - |\rho|^2}}\int_0^\infty f_{|U_1|^2}(s)\, f_{|U_2|^2}\left(s - \frac{2t_0}{\sqrt{1 - |\rho|^2}}\right) ds \tag{29}$$

in which $|U_i|^2 \sim \chi_2^2(\kappa_i)$ represent noncentral chi-squared distributed random variables with noncentrality parameters $\kappa_i = |a_0|^2\boldsymbol{S}_0^H\mathbf{R}^{-1}\boldsymbol{S}_0(1 + (-1)^i\sqrt{1 - |\rho|^2})/2$ for $i = 1, 2$. The corresponding cumulative density function (cdf) $F(\cdot)$ can be computed via

$$F_{T_0}(t_0) = 1 - \int_0^\infty f_{|U_1|^2}(s)\, F_{|U_2|^2}\left(s - \frac{2t_0}{\sqrt{1 - |\rho|^2}}\right) ds. \tag{30}$$

The pdf for the case $K > 1$ can easily be derived using the fact that the sum of mutually independent noncentral chi-squared distributed random variables has a distribution of the same type with $2K$ degrees of freedom and its noncentrality parameters are the sum of the corresponding parameters of all the summands.

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